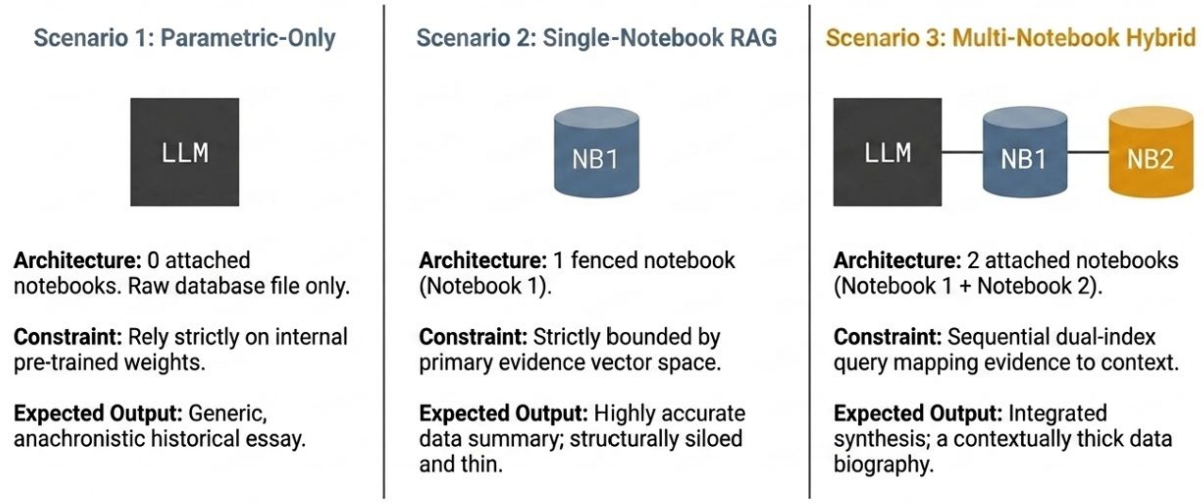
Phase 3: Scenario Execution (The Generative Process)
Overview of Experimental Design

In Phase 3, identical, unified experimental prompts to generate the SDB are executed under a comparative three condition design.Three experimental conditions are constructed to isolate the contribution of contextual retrieval and multi-notebook synthesis. The foundational LLM, generation prompts, and evaluation procedures are held strictly constant across all conditions. This design isolates retrieval architecture as the sole manipulated variable.

- **Scenario 1 (Parametric Baseline):** A standalone large language model is provided with only the raw database file and no external retrieval index, establishing the generative ceiling of parametric knowledge alone. This scenario serves as the experimental control
- **Scenario 2 (Single-Notebook RAG):** A (fenced) single-notebook RAG configuration in which the model retrieves exclusively from a primary corpus notebook containing the datafied records, fencing the system from external historical scholarship.
- **Scenario 3 (Multi-Notebook Hybrid RAG):** A dual-notebook RAG configuration in which the model retrieves sequentially from both a primary corpus notebook and an curated context notebook containing historical, legal, institutional, and cultural knowledge relevant to the conditions of original information generation. The architecture was instructed to synthesize evidence retrieved from both repositories while avoiding unsupported interpolation from parametric knowledge.

Figure 1: Comparative Three-Condition Design



Independent and Dependent Variables. The independent variable is the **retrieval architecture** used to generate the sociocultural data biography. It is operationalized through three experimental conditions that provide progressively richer access to external knowledge: (1) parametric generation without notebook-based retrieval; (2) retrieval from the primary corpus (Notebook 1); and (3) retrieval from both the primary corpus and the sociocultural context notebook (Notebook 2)

The three experimental conditions represent alternative retrieval architectures rather than successive levels of a single quantitative variable. Because they differ in certain implementation characteristics (e.g. direct-ingestion-versus-indexed-retrieval mechanism), pairwise comparisons should be interpreted accordingly. In particular, the comparison between Conditions 2 and 3 provides the clearest assessment of the contribution of the sociocultural context notebook, whereas comparisons involving Condition 1 compare complete retrieval configurations rather than isolating individual implementation features.

This comparative experiment evaluates two distinct dependent variables, corresponding to RQ2a and RQ2b respectively, and does not treat them as components of a single composite measure.

- Technical RAG performance, operationalized through an adapted RAG Triad (faithfulness, context recall, answer relevance), addressing RQ2a.
- Methodological adequacy of the generated SDB, operationalized through the four adequacy criteria (contextual, evidentiary, translational, analytical utility), addressing RQ2b.

These two dependent variables are evaluated independently by separate judge prompts, separate evaluation rubrics, and separate LLM judge models. The adapted RAG Triad evaluates the technical performance of the retrieval architecture, whereas the SDB adequacy framework evaluates the methodological adequacy of the generated sociocultural data biography. The two evaluations are reported separately and are not combined into a composite score because they address different research questions and evaluate different aspects of system performance.

The unit of analysis is the SDB generated under a single experimental condition, evaluated once by each of three judge models per dependent variable.

Experimental Expectations. As this is an exploratory study, the empirical outcome of interest is not hypothesis confirmation but whether the three retrieval conditions produce meaningfully different SDBs, and whether those differences are visible in both the RAG metrics and the SDB adequacy criteria. The most informative outcome would be a pattern in which the dual-notebook condition (Scenario 3) outperforms the other two on contextual recall, evidentiary grounding, translational guidance, and analytical utility, while the parametric-only and single-notebook conditions produce "thinner" or less contextually grounded biographies. Such a result would not prove the framework universally, but it would support the claim that added sociocultural context changes the quality of the generated artifact in the expected direction.

Experimental Technical Environment. To maintain experimental consistency, structural alignment, and strict environmental control across all three experimental conditions, the experiment was conducted primarily within the Google ecosystem. Two software environments were employed: a standalone **NotebookLM** workspace for the fenced retrieval configurations (Scenario 2) and pre-test audits; and the **Gemini** interface for the parametric control and multi-notebook synthesis runs (Scenarios 1 and 3, respectively). For evaluation, LLM-as-judge was

implemented for each evaluation tool, across three LLMs (Gemini, ChatGPT and Claude) to provide cross-platform comparisons of consistency in ratings.

Using a common software ecosystem reduced methodological variability that could have been introduced through differences among commercial RAG platforms. Because all three experimental conditions were implemented using closely related software environments, observed differences among the AI-generated SDBs are more plausibly attributable to the retrieval configurations under investigation than to differences in platform implementation.

The software environment also supported experimental design in several practical ways:

- The primary corpus and the contextual knowledge base could be ingested without requiring subdivision or restructuring of the experimental materials prior to generation.
- NotebookLM and Gemini provided distinct retrieval environments that enabled the experimental conditions to be implemented while maintaining clear separation between the primary corpus and the contextual knowledge base. This separation was essential both for the KBS pre-test audit, in which the two knowledge repositories were evaluated independently, and for the dual-notebook retrieval condition, in which the repositories were queried together.
- Each experimental condition was conducted in a newly initiated and contained session. Beginning each scenario in a separate session prevented information generated during earlier runs from remaining available within the active conversational context, thereby reducing the possibility of contextual carryover between experimental conditions.

The decision to conduct the experiment within a single software ecosystem necessarily limits direct comparison with RAG systems implemented on other platforms. However, this design choice strengthens the study's internal validity by reducing implementation differences unrelated to the retrieval architectures being evaluated.

Table 1: System constraint instructions

Scenario 1(Control)	**Scenario 2 (Fenced Context)**	**Scenario 3 (Integrated synthesis)**
System constraint added to uniform prompt: *"Generate this biography utilizing only your internal parametric weights and the raw database fields provided in the attached file. Do not attempt to dynamically infer outside academic frameworks or replicate unprovided metadata*	System constraint added to uniform prompt: "Your generation must strictly be bounded by the semantic vector space of this notebook. If an interpretive macro-pattern cannot be directly proven or cited using the explicit data strings in this index, you must declare it unverified rather than synthesizing general historical filler."	System constraint added to uniform prompt: *"Execute a sequential, dual-index query. First, map the macro-level transactional patterns across the fields of Notebook 1. Second, explicitly cross-reference those evidence points with the corresponding legal and cultural notes in Notebook 2. You are strictly forbidden from using generalized parametric tropes to bridge gaps between these two independent indexes."*

To avoid contextual carryover between conditions, each scenario was conducted in a newly initiated session. The AI-generated SDBs subsequently are evaluated using identical LLM-as-judge instruments and scoring rubrics. This design isolates retrieval architecture as the only manipulated variable, allowing observed differences among the generated biographies to be attributed to differences in contextual retrieval rather than changes in prompts, corpus, or evaluation procedures.

Uniform SDB generation prompt. A single uniform generation prompt was developed to produce sociocultural data biographies (SDBs) under each experimental condition (Scenarios 1 – 3). The prompt functions as a controlled research instrument and remains constant throughout formal data collection. A scenario-specific retrieval instruction ("system constraint") is added to the uniform prompt for each of the three experimental conditions to define the conditions for information retrieval (Table 1). Maintaining a uniform prompt ensures that observed differences among generated sociocultural data biographies reflected differences in retrieval architecture rather than differences in task formulation or prompting strategy.

The uniform prompt is designed to operationalize the conceptual definition of SDB. To ensure consistency across experimental conditions, the prompt incorporates several common components:

1. It provides the formal conceptual definition of the SDB, which served as the theoretical foundation for the experiment.
2. It establishes explicit boundaries governing the dataset schema by identifying the twenty-seven (27) verified fields that constituted the primary corpus (Notebook 1) and instructing the model not to infer additional variables beyond those present in the dataset.
3. It requires that all corpus-level claims be explicitly limited to the primary corpus unless independently supported by contextual evidence.
4. It specifies a common report structure corresponding to the four adequacy criteria evaluated in this study: contextual adequacy, evidentiary adequacy, translational adequacy, and analytical utility.

The text of the uniform SDB generation prompt is provided in this repository. The architecture of the uniform prompt is shown in Figure 2

Each section of the required report contains explicit instructions describing the analytical content expected in the AI-generated SDB. These instructions operationalize the SDB conceptual framework by directing the model to document contextual conditions, identify evidentiary support, translate contextual findings into methodological guidance, and demonstrate the interpretive value of the resulting SDB throughout the computational research lifecycle.

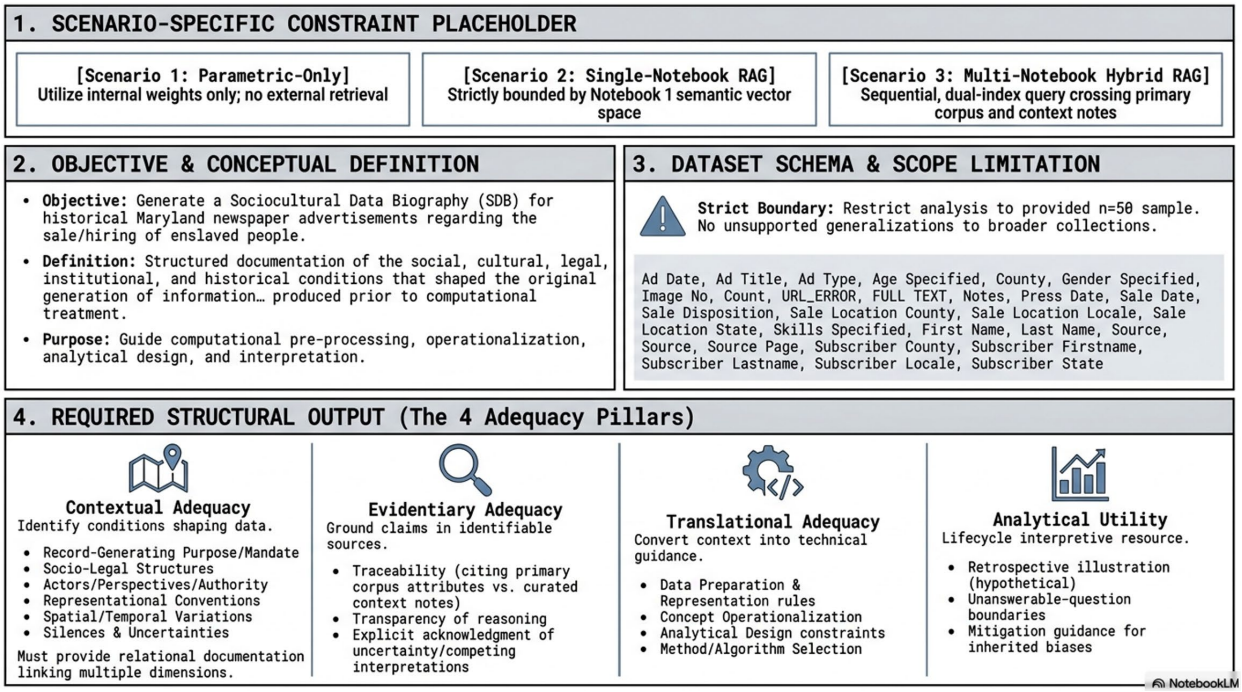


Fig. 2: Architecture of the Uniform SDB Generation Prompt

Each section of the required report contains explicit instructions describing the analytical content expected in the AI-generated SDB. These instructions operationalize the SDB conceptual framework by directing the model to document contextual conditions, identify evidentiary support, translate contextual findings into methodological guidance, and demonstrate the interpretive value of the resulting SDB throughout the computational research lifecycle.

Scenario Prompt Constraints. The only instruction that varies by scenario is the single system-constraint sentence inserted at the top of the prompt (Table 1). These three constraint sentences remained unchanged through pilot versions of the prompt, as they govern retrieval architecture rather than retrieval performance or adequacy-criteria content. They constituted the only substantive differences among the generation prompts used during formal experimentation.

Output Requirements. The prompt requires a four-section structured report based on the four SDB adequacy criteria: contextual adequacy, evidentiary adequacy, translational adequacy, and analytic utility. Contextual adequacy is operationalized (Section 4.6) to address six analytical dimensions (record-generating purpose and institutional mandate; socio-legal structures; actors, perspectives, and authority; representational conventions; spatial and temporal variation; and structured absences, silences, and uncertainties), plus at least one explicit account of intersecting

dimensions producing a feature no single dimension explains alone. Evidentiary adequacy is operationalized to ground every contextual claim in traceable evidence, with explicit reasoning transparency, uncertainty acknowledgement, and a citation-specificity requirement. Translational adequacy translates documented context into guidance on four decision categories (data preparation and representation; concept operationalization; analytical design; method and algorithm selection), each justified by a specific contextual finding, with an ambiguity clause restricted to genuine forks in computational treatment rather than data-reliability questions. Analytic utility demonstrates the SDB's retrospective interpretive function through an explicitly hypothetical illustration, identifying at least one category of analytical question the corpus cannot reliably answer, and avoiding redundancy.

Controls and Consistency. The citation-specificity instruction in Evidentiary Adequacy section of the output requirements was applied uniformly across all three scenarios rather than conditionally, even though there is no Notebook 2 to cite in Scenarios 1 and 2. The uniform application design feature emerged during pilot testing as an instrument-consistency decision. In Scenario 1 and 2, it functions as a negative-control check, such that any appearance of an author/title-style citation in Scenario 1 or 2 output, despite the absence of a context notebook, is treated as a specific, diagnostic finding of parametric-knowledge leakage rather than as successful compliance.

Pilot Testing and Schema Correction. The generation prompt underwent iterative refinement during pilot testing. Pilot evaluation identified a schema-fidelity error resulting from inconsistencies between the prompt and the verified structure of primary corpus. Specifically, one dataset field was referenced under an incorrect name, and three additional fields were identified as inconsistent with the corrected twenty-seven-field schema.

Because the SDB generation prompt instructed the model to restrict its analysis strictly to the field names provided, these discrepancies meant that the pilot-phase prompt directed the generating model toward one nonexistent field and three misnamed fields, independent of any property of the model's reasoning or retrieval behavior.

This schema-fidelity error was identified only after pilot-phase Scenario 1 and Scenario 2 outputs had already been generated under the uncorrected prompt. Rather than retain this data as

part of the study's reported findings, the pilot-phase outputs for both scenarios were formally set aside as calibration data. This decision was made on the grounds that any output generated under a prompt referencing an incorrect schema could not be treated as representative of the generating model's performance under the experimental conditions the study was designed to test; retaining such outputs in the reported comparison would risk conflating instrument error with model or architectural performance.

Following this determination, the generation prompt was revised to reference the verified, twenty-seven-field schema of the primary corpus. The nonexistent Ad Date reference was corrected by renaming the source column from Column 1 to Ad Date in the dataset itself, so that the field genuinely existed under the name the prompt required; and the three misnamed fields were corrected to Age Specified, Gender Specified, and Subscriber Firstname, respectively. The dataset documentation and the operational definitions of the four adequacy criteria, both of which had been drafted with reference to an earlier and likewise uncorrected schema, were subsequently revised to maintain consistency with the corrected field names used in the generation prompt. All results reported in this study were generated under the corrected prompt. No pilot-phase output is included in the reported comparison across Scenarios 1 through 3.